The Conditional Activation Model: Uncovering Policy Demand Responses to AI-Driven Automation

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March 9, 2026

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Outline

1. Puzzle & Theory
2. Research Design
3. Identification & Model
4. Results
5. Discussion

Puzzle & Theory

The Automation–Policy Demand Puzzle

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$\Rightarrow$ **When—and for whom—does exposure to automation risk translate into policy demand?**

Mixed Findings in the Literature

Study	Trust Modeled?	Prior Beliefs Elicited?	Engagement Required?	Policy Framing	Finding
Zhang (2022)	No	No	No (vignette)	Mixed	Null
Magistro et al. (2024)	No	No	No (conjoint)	Contextualized	Null
Ladreit (2022)	No	No	No (vignette)	Contextualized	Null
Arntz et al. (2024)	No	Yes (optimism)	No	Traditional	**Heterogeneous**
Heinrich & Witko (2024)	Proxy (knowledge)	No	No	Mixed	**Conditional+**
Haslberger (2024)	Reduced (behavioral)	No	Yes (AI use)	Traditional	**Positive**
This Study	**Yes (latent)**	**Yes (Gap)**	**Yes (3-step)**	**Contextualized**	**CATE**

Studies without trust modeling or belief elicitation $\rightarrow$ null effects.

Design features accumulate $\rightarrow$ conditional effects emerge.

Theoretical Ingredients

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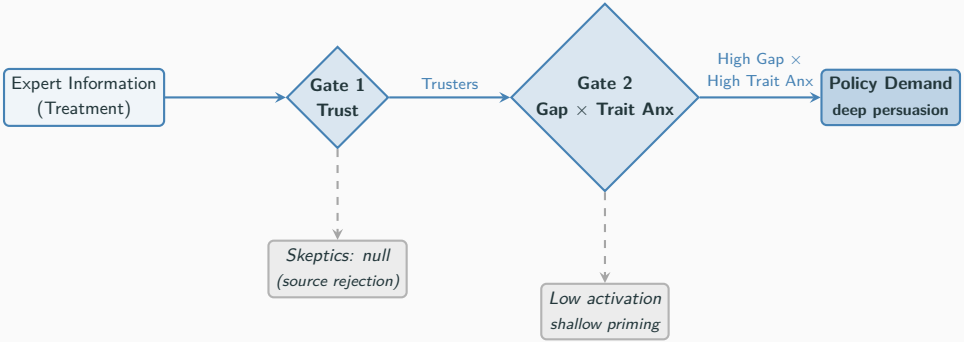
- High-disposition individuals override partisan heuristics and engage effortfully with threatening information *(Albertson & Gadarian, 2015)*.
- **Belief Gap** = expert estimate − prior belief: the "activation potential" for updating.

The Conditional Activation Model



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H5. Conditional Activation. Largest effects among Trusters with high anxiety *and* positive Gaps—both gates satisfied.

The Right Estimand

Why the ATE understates the story:

$$\text{ATE} = \underbrace{\pi}_{\text{Truster share}} \cdot \tau_{\mathrm{T}} + \underbrace{(1-\pi)}_{\text{Skeptic share}} \cdot \underbrace{\tau_{\mathrm{S}}}_{\approx 0}$$

If $\tau_{\mathrm{S}} \approx 0$, the ATE **dilutes** the conditional signal. The theoretically relevant quantity is the **CATE**.

Research Design

Sample and Randomization Balance

Sample

- $N = 901$ U.S. workers via CloudResearch Connect
- Targeting (exclusion enforced): ages 18–65; full-time, part-time, business owner, or unemployed; English-speaking
- Treatment: $n = 475$ Control: $n = 426$
- Between-subjects randomization

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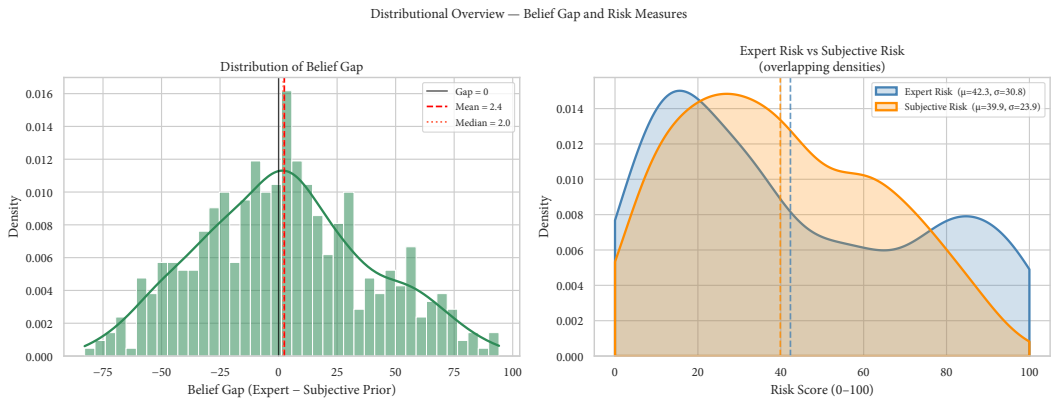
Manipulation checks

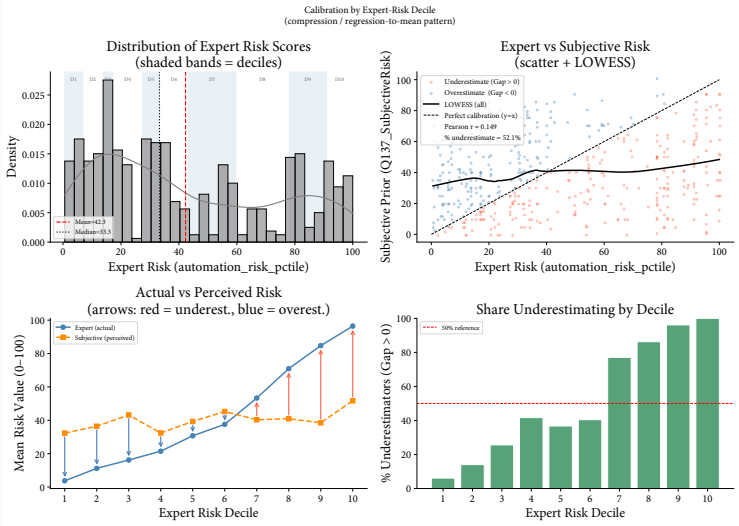
- 93.4% comprehension on treatment text
- Perceived gap tracks objective gap ($\rho = 0.73$, $p < .001$)
- Belief updating tracks gap magnitude ($\rho = 0.66$, $p < .001$)

Treatment: Three-Step Belief Correction

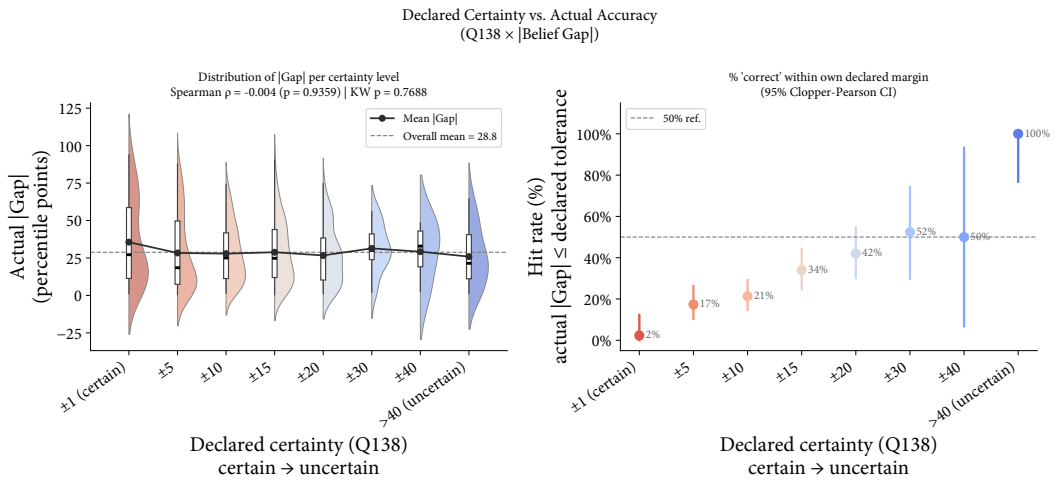
The Belief Gap: Construction and Distribution



Belief Gap: Systematic Miscalibration



Belief Gap: Workers Don't Know How Wrong They Are

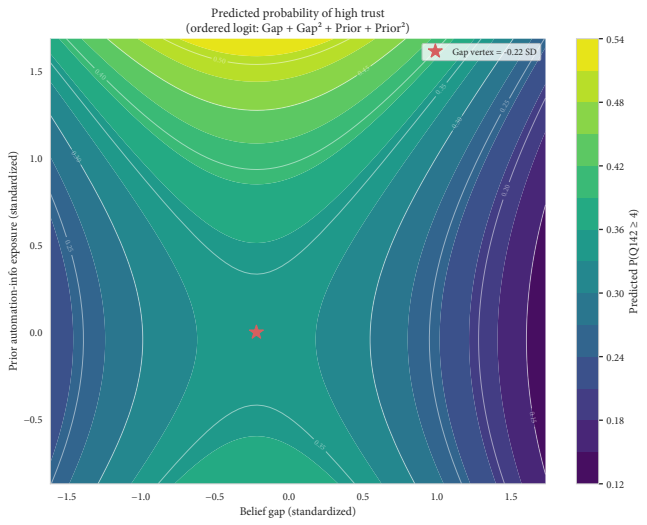


Manipulation Checks

Identification & Model

The Identification Problem: Trust as a Collider

Trust Covariates: Epistemic Validation

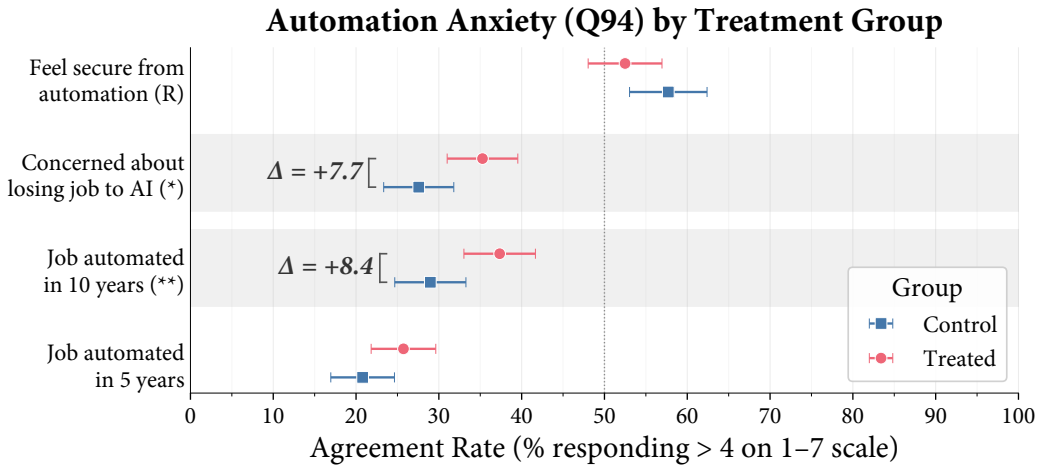


Bayesian Finite Mixture Model

Results

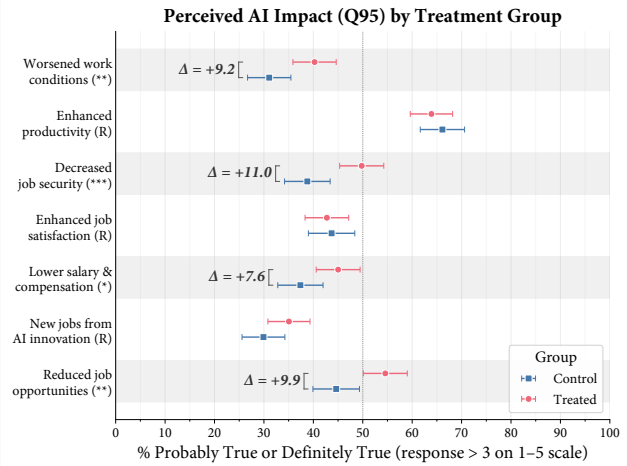
Unconditional Average Treatment Effect

Item-Level Effects: Automation Anxiety (Q94)

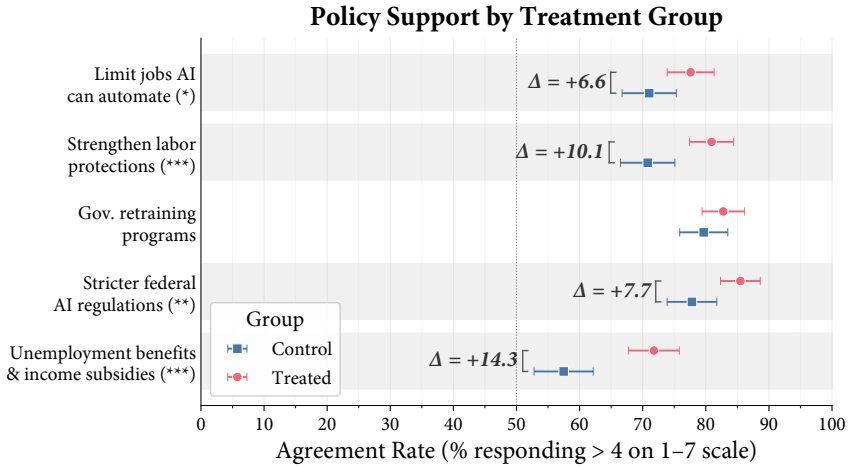


Note: * p < .05, ** p < .01, *** p < .001 (two-proportion z-test). Shaded rows = p < .05.
Δ = Treated − Control (pp). Error bars = 95% Wald CI. (R) = reverse-worded item.

Item-Level Effects: Perceived AI Impact (Q95)



Item-Level Effects: Policy Support (Q96)



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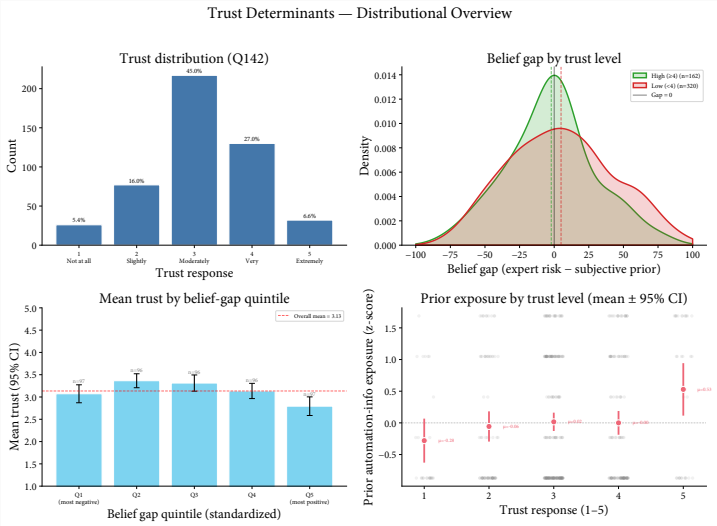
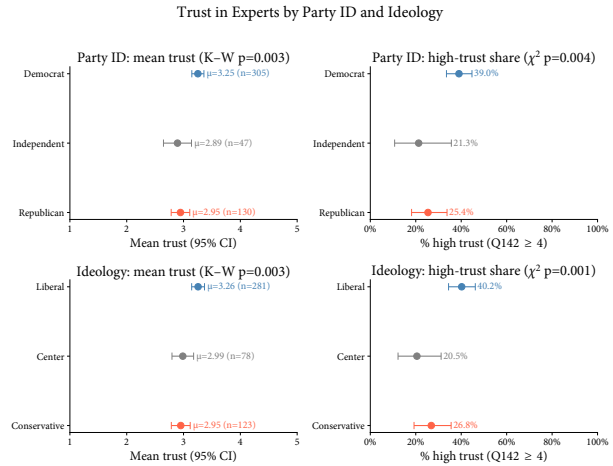
First Stage: Latent Trust Classes



Trust and Partisan Identity

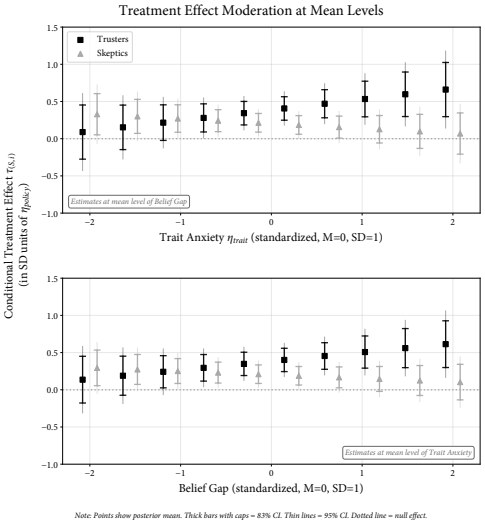


Note: K–W = Kruskal–Wallis H-test, a non-parametric one-way ANOVA that tests whether the trust distributions differ across groups without assuming normality or equal variances.
χ^2 = Pearson chi-squared test for independence of the high-trust binary indicator.

CATE Surface: Treatment Effects Among Trusters

CATE Contrasts: Maximum Activation Dominates

Partial Effects: Gate 1 Is a Strict Filter



Discussion

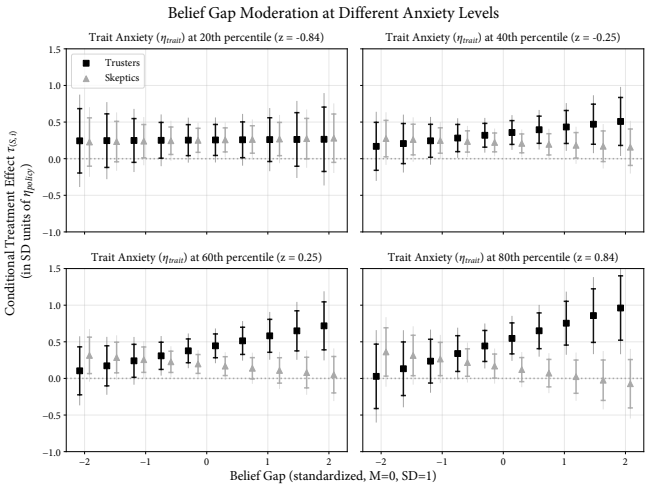
Three Pathways, Not Two

Limitations

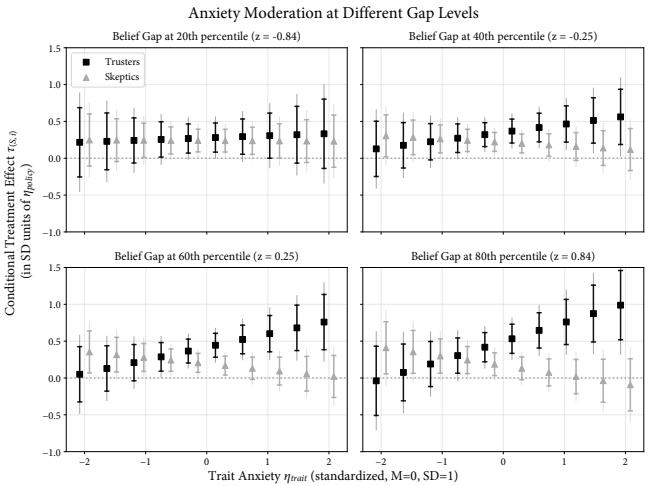
Takeaway

Appendix

Gap Slope Steepens With Anxiety



Anxiety Slope Steepens With Belief Gap



Note: Points show posterior mean. Thick bars with caps = 83% CI. Thin lines = 95% CI. Dotted line = null effect.

Model Coefficients (Forest Plot)

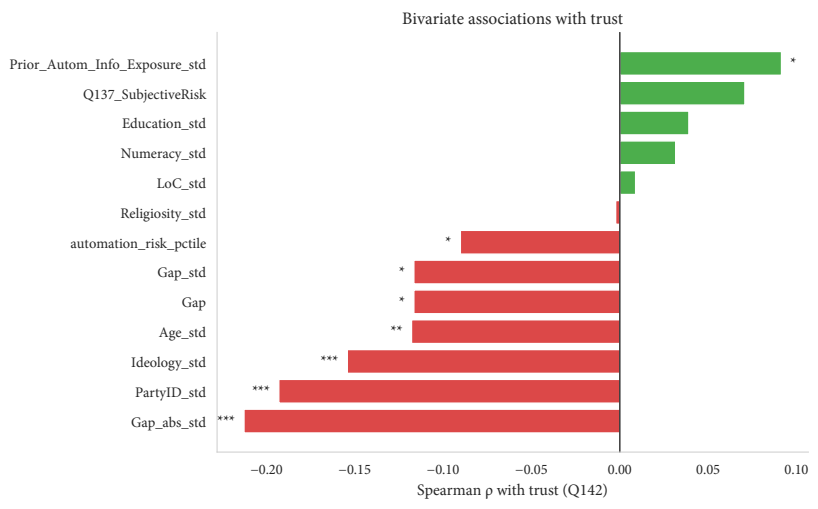
BFMM Treatment Effects (Full Table)

Partisan Heterogeneity

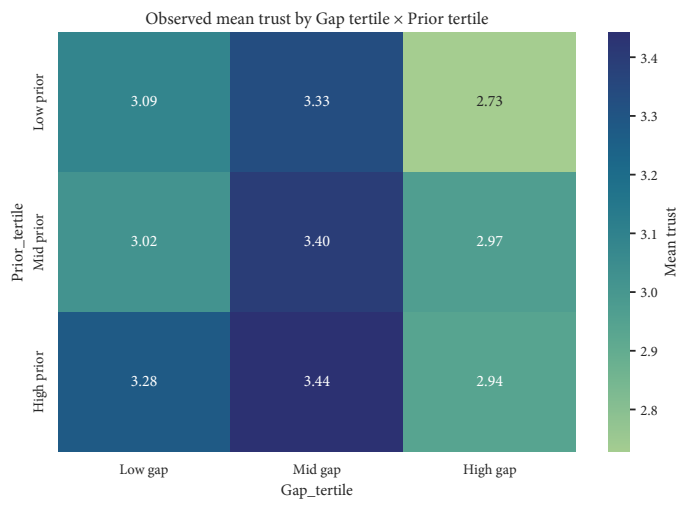
Mediation Analysis

Scale Reliability

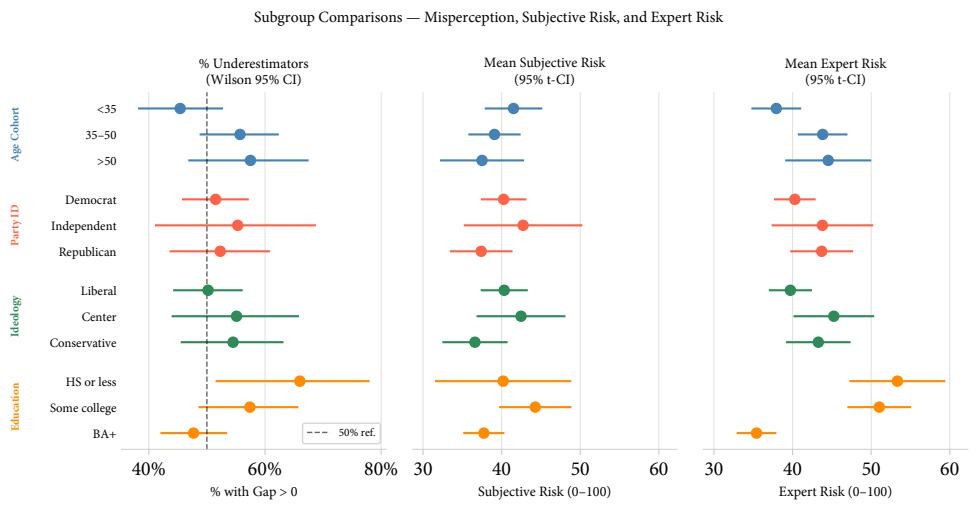
Trust Correlates



Trust by Gap × Prior Exposure (Raw Means)



Belief Gap Across Demographic Subgroups



Prior Distributions

Sample Demographics

Descriptive Statistics

Full Balance — Continuous Covariates

Full Balance — Categorical Covariates

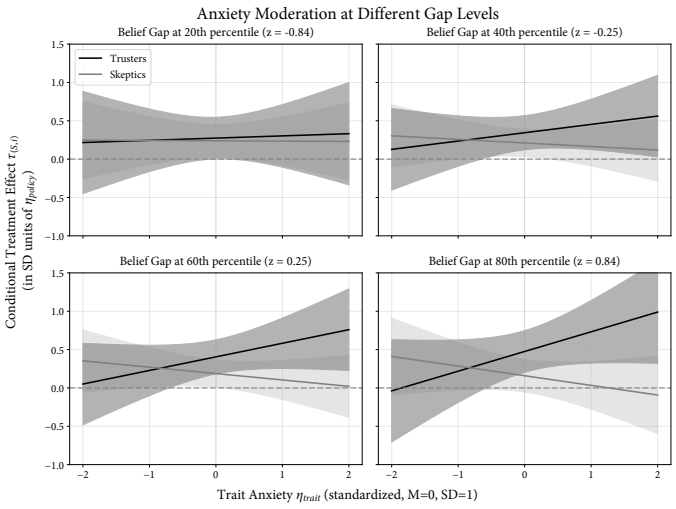
Scale Inter-Correlations

Trust Mixture Covariates

What Predicts the Belief Gap?

Full Manipulation Checks

Anxiety Partial Effects (Continuous Ribbons)



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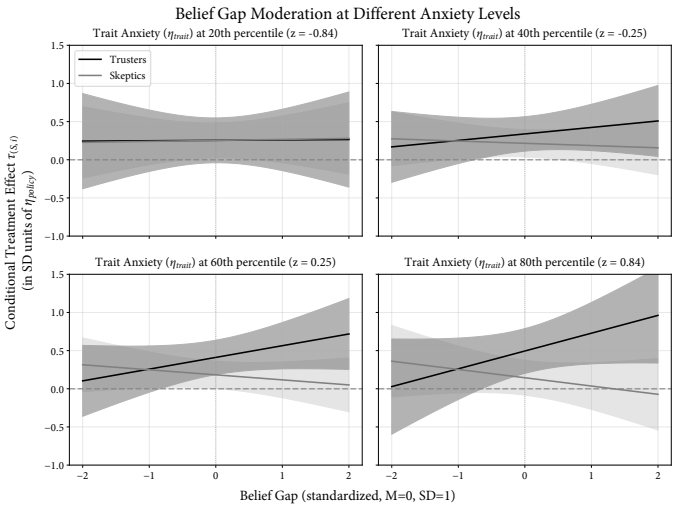
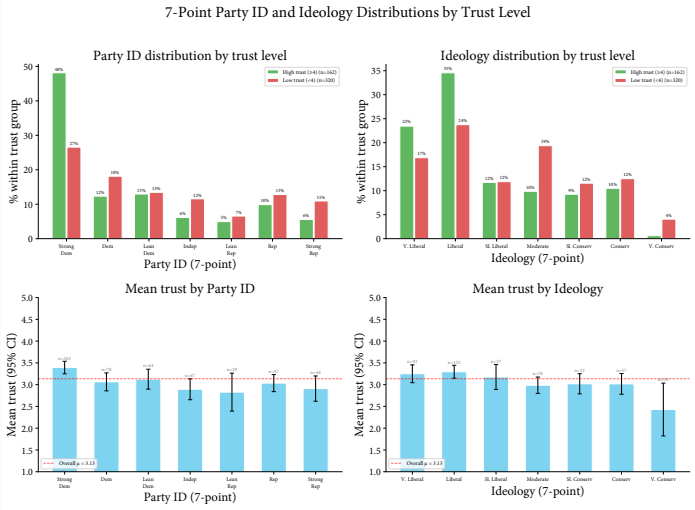


Single-Moderator Partial Effects

Trust by 7-Point Party ID and Ideology



Automation Anxiety Item Effects

Perceived AI Impact Item Effects

Full Mediation Path Decomposition

Treatment Effects on Mechanism Scales

Partisan Effects vs. Pre-Registered Criteria

Partisan × Trust Heterogeneity